

Лабораторная Работа №16. Настройка VPN

Администрирование локальных сетей

Боровиков Д.А.

Российский университет дружбы народов им. Патриса Лумумбы, Москва, Россия

- Боровиков Даниил Александрович
- НПИБд-01-22
- Российский университет дружбы народов
- [1132222006@pfur.ru]

Получение навыков настройки VPN-туннеля через незащищённое Интернет-соединение.

Размещение оборудования

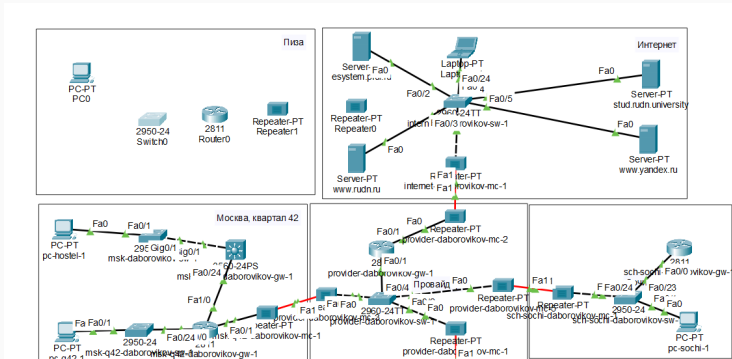


Figure 1: Размещение оборудования в рабочей области проекта.



Figure 2: Замена модулей на Repeater-PT

Подключение оборудования

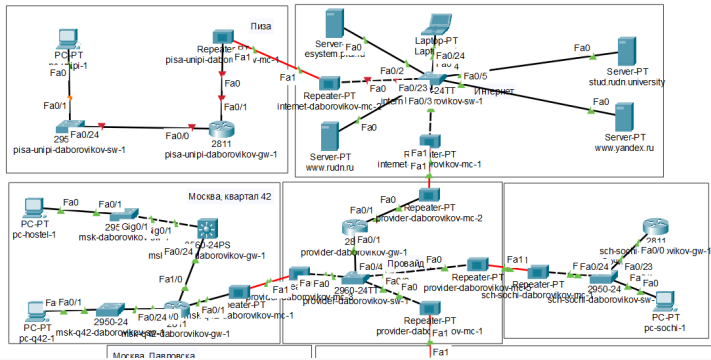


Figure 3: Подключение оборудования

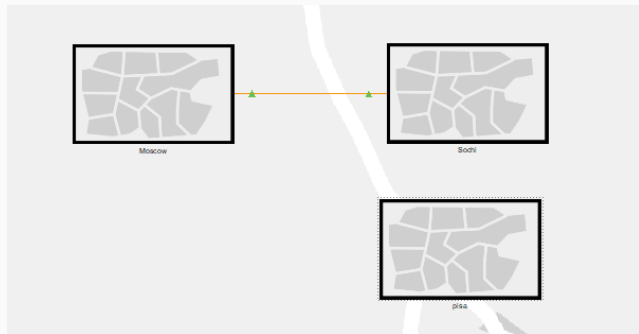


Figure 4: Создание города Пиза в физической рабочей области

Перемещение оборудования

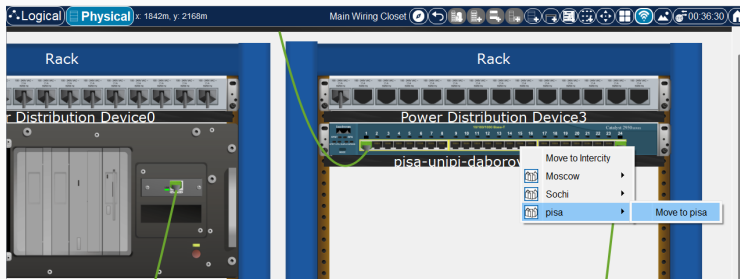
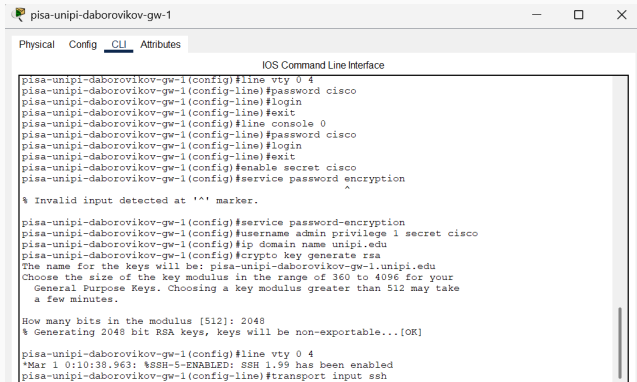


Figure 5: Перемещение оборудования



The screenshot shows a window titled "pisa-unipi-daborovikov-gw-1" with tabs for "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is active, displaying the "IOS Command Line Interface". The interface shows the following commands and their outputs:

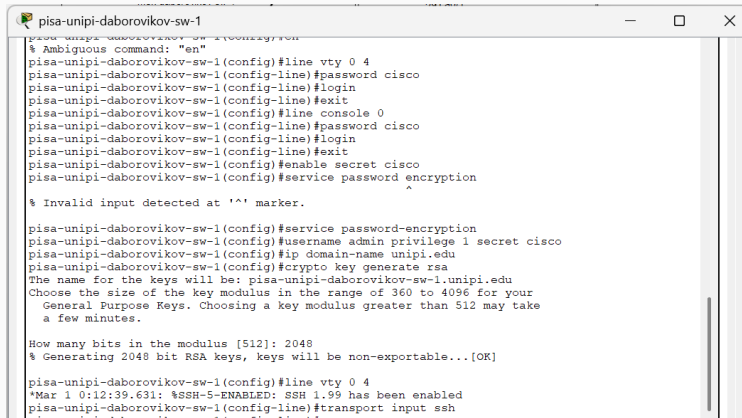
```
pisa-unipi-daborovikov-gw-1(config)#line vty 0 4
pisa-unipi-daborovikov-gw-1(config-line)#password cisco
pisa-unipi-daborovikov-gw-1(config-line)#login
pisa-unipi-daborovikov-gw-1(config-line)#exit
pisa-unipi-daborovikov-gw-1(config)#line console 0
pisa-unipi-daborovikov-gw-1(config-line)#password cisco
pisa-unipi-daborovikov-gw-1(config-line)#login
pisa-unipi-daborovikov-gw-1(config-line)#exit
pisa-unipi-daborovikov-gw-1(config)#enable secret cisco
pisa-unipi-daborovikov-gw-1(config)#service password encryption
^
% Invalid input detected at '^' marker.

pisa-unipi-daborovikov-gw-1(config)#service password-encryption
pisa-unipi-daborovikov-gw-1(config)#username admin privilege 1 secret cisco
pisa-unipi-daborovikov-gw-1(config)#ip domain name unipi.edu
pisa-unipi-daborovikov-gw-1(config)#crypto key generate rsa
The name for the keys will be: pisa-unipi-daborovikov-gw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-daborovikov-gw-1(config)#line vty 0 4
*Mar 1 0:10:38.963: %SSH-5-ENABLED: SSH 1.99 has been enabled
pisa-unipi-daborovikov-gw-1(config-line)#transport input ssh
```

Figure 6: Первоначальная настройка маршрутизатора pisa-unipi-daborovikov-gw-1



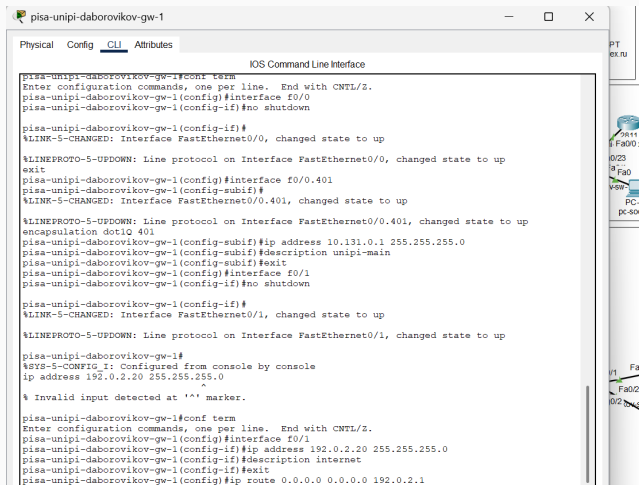
```
pisa-unipi-daborovikov-sw-1
pisa-unipi-daborovikov-sw-1(config)#
% Ambiguous command: "en"
pisa-unipi-daborovikov-sw-1(config)#line vty 0 4
pisa-unipi-daborovikov-sw-1(config-line)#password cisco
pisa-unipi-daborovikov-sw-1(config-line)#login
pisa-unipi-daborovikov-sw-1(config-line)#exit
pisa-unipi-daborovikov-sw-1(config)#line console 0
pisa-unipi-daborovikov-sw-1(config-line)#password cisco
pisa-unipi-daborovikov-sw-1(config-line)#login
pisa-unipi-daborovikov-sw-1(config-line)#exit
pisa-unipi-daborovikov-sw-1(config)#enable secret cisco
pisa-unipi-daborovikov-sw-1(config)#service password encryption
^
% Invalid input detected at '^' marker.

pisa-unipi-daborovikov-sw-1(config)#service password-encryption
pisa-unipi-daborovikov-sw-1(config)#username admin privilege 1 secret cisco
pisa-unipi-daborovikov-sw-1(config)#ip domain-name unipi.edu
pisa-unipi-daborovikov-sw-1(config)#crypto key generate rsa
The name for the keys will be: pisa-unipi-daborovikov-sw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

pisa-unipi-daborovikov-sw-1(config)#line vty 0 4
*Mar 1 0:12:39.631: %SSH-5-ENABLED: SSH 1.99 has been enabled
pisa-unipi-daborovikov-sw-1(config-line)#transport input ssh
```

Figure 7: Первоначальная настройка коммутатора pisa-unipi-daborovikov-sw-1



```
pisa-unipi-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
pisa-unipi-daborovikov-gw-1(config)#interface f0/0
pisa-unipi-daborovikov-gw-1(config-if)#no shutdown

pisa-unipi-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
pisa-unipi-daborovikov-gw-1(config)#interface f0/0.401
pisa-unipi-daborovikov-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up
encapsulation dot1Q 401
pisa-unipi-daborovikov-gw-1(config-subif)#ip address 10.131.0.1 255.255.255.0
pisa-unipi-daborovikov-gw-1(config-subif)#description unipi-main
pisa-unipi-daborovikov-gw-1(config-subif)#exit
pisa-unipi-daborovikov-gw-1(config)#interface f0/1
pisa-unipi-daborovikov-gw-1(config-if)#no shutdown

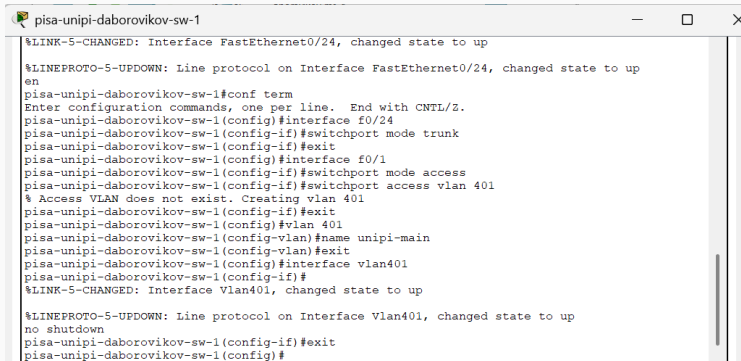
pisa-unipi-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

pisa-unipi-daborovikov-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
ip address 192.0.2.20 255.255.255.0
^
% Invalid input detected at '^' marker.

pisa-unipi-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
pisa-unipi-daborovikov-gw-1(config)#interface f0/1
pisa-unipi-daborovikov-gw-1(config-if)#ip address 192.0.2.20 255.255.255.0
pisa-unipi-daborovikov-gw-1(config-if)#description internet
pisa-unipi-daborovikov-gw-1(config-if)#exit
pisa-unipi-daborovikov-gw-1(config)#ip route 0.0.0.0 0.0.0.0 192.0.2.1
```

Figure 8: Настройка интерфейсов маршрутизатора pisa-unipi-daborovikov-gw-1



```
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
en
pisa-unipi-daborovikov-sw-1#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
pisa-unipi-daborovikov-sw-1(config)#interface f0/24
pisa-unipi-daborovikov-sw-1(config-if)#switchport mode trunk
pisa-unipi-daborovikov-sw-1(config-if)#exit
pisa-unipi-daborovikov-sw-1(config)#interface f0/1
pisa-unipi-daborovikov-sw-1(config-if)#switchport mode access
pisa-unipi-daborovikov-sw-1(config-if)#switchport access vlan 401
% Access VLAN does not exist. Creating vlan 401
pisa-unipi-daborovikov-sw-1(config-if)#exit
pisa-unipi-daborovikov-sw-1(config)#vlan 401
pisa-unipi-daborovikov-sw-1(config-vlan)#name unipi-main
pisa-unipi-daborovikov-sw-1(config-vlan)#exit
pisa-unipi-daborovikov-sw-1(config)#interface vlan401
pisa-unipi-daborovikov-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up
no shutdown
pisa-unipi-daborovikov-sw-1(config-if)#exit
pisa-unipi-daborovikov-sw-1(config)#
```

Figure 9: Настройка интерфейсов коммутатора pisa-unipi-daborovikov-sw-1

Присвоение адресов

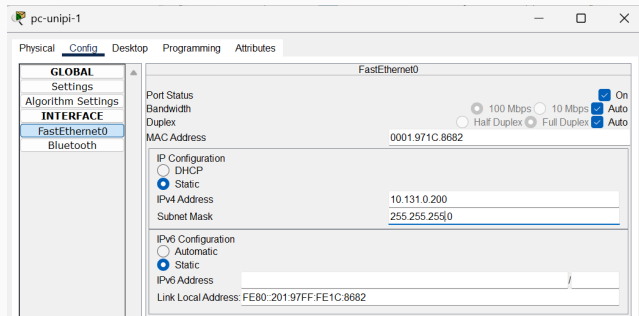
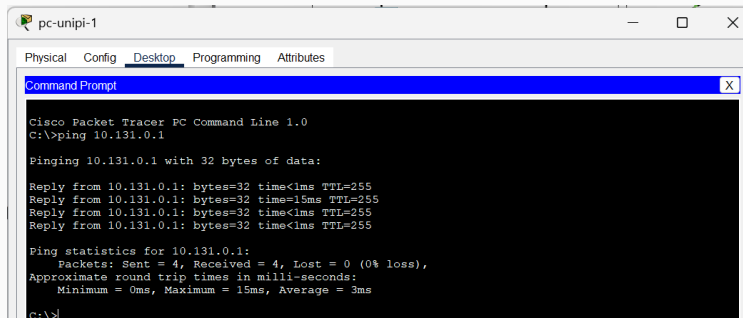


Figure 10: Присвоение адресов оконечному устройству



The screenshot shows a window titled "pc-unipi-1" with tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is active, displaying a "Command Prompt" window. The command prompt shows the execution of the command "ping 10.131.0.1" and its output, which includes four successful replies and summary statistics.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.131.0.1

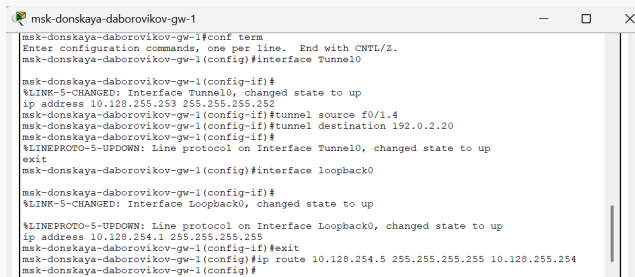
Pinging 10.131.0.1 with 32 bytes of data:

Reply from 10.131.0.1: bytes=32 time<1ms TTL=255
Reply from 10.131.0.1: bytes=32 time=15ms TTL=255
Reply from 10.131.0.1: bytes=32 time<1ms TTL=255
Reply from 10.131.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.131.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 15ms, Average = 3ms

C:\>
```

Figure 11: Пинг адреса 10.131.0.1



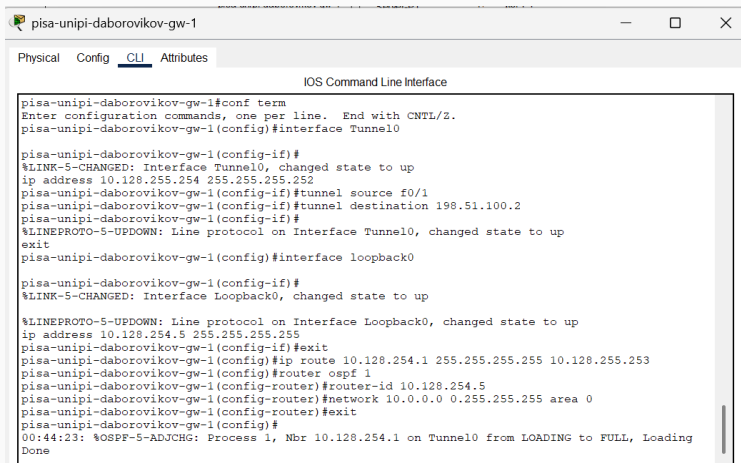
```
msk-donskaya-daborovikov-gw-1
msk-donskaya-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-daborovikov-gw-1(config)#interface Tunnel0

msk-donskaya-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up
ip address 10.128.255.253 255.255.255.252
msk-donskaya-daborovikov-gw-1(config-if)#tunnel source f0/1.4
msk-donskaya-daborovikov-gw-1(config-if)#tunnel destination 192.0.2.20
msk-donskaya-daborovikov-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up
exit
msk-donskaya-daborovikov-gw-1(config)#interface loopback0

msk-donskaya-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
ip address 10.128.254.1 255.255.255.255
msk-donskaya-daborovikov-gw-1(config-if)#exit
msk-donskaya-daborovikov-gw-1(config)#ip route 10.128.254.5 255.255.255.255 10.128.255.254
msk-donskaya-daborovikov-gw-1(config)#
```

Figure 12: Настройка маршрутизатора msk-donskaya-daborovikov-gw-1



```
pisa-unipi-daborovikov-gw-1
Physical Config CLI Attributes
IOS Command Line Interface

pisa-unipi-daborovikov-gw-1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
pisa-unipi-daborovikov-gw-1(config)#interface Tunnel0

pisa-unipi-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up
ip address 10.128.255.254 255.255.255.252
pisa-unipi-daborovikov-gw-1(config-if)#tunnel source f0/1
pisa-unipi-daborovikov-gw-1(config-if)#tunnel destination 198.51.100.2
pisa-unipi-daborovikov-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up
exit
pisa-unipi-daborovikov-gw-1(config)#interface loopback0

pisa-unipi-daborovikov-gw-1(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
ip address 10.128.254.5 255.255.255.255
pisa-unipi-daborovikov-gw-1(config-if)#exit
pisa-unipi-daborovikov-gw-1(config)#ip route 10.128.254.1 255.255.255.255 10.128.255.253
pisa-unipi-daborovikov-gw-1(config)#router ospf 1
pisa-unipi-daborovikov-gw-1(config-router)#router-id 10.128.254.5
pisa-unipi-daborovikov-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
pisa-unipi-daborovikov-gw-1(config-router)#exit
pisa-unipi-daborovikov-gw-1(config)#
00:44:23: %OSPF-5-ADJCHG: Process 1, Nbr 10.128.254.1 on Tunnel0 from LOADING to FULL, Loading
Done
```

Figure 13: Настройка маршрутизатора pisa-unipi-daborovikov-gw-1

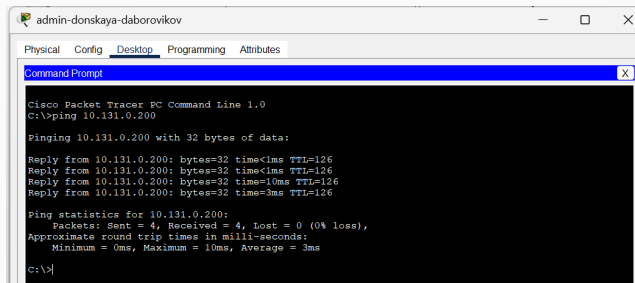


Figure 14: Проверка доступности узлов сети Университета г. Пиза с ноутбука администратора сети «Донская»

В ходе выполнения лабораторной работы мы получили навыки настройки VPN-туннеля через незащищённое Интернет-соединение.